



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

Digital Forensics

NIST Hacking Case

Scenario

- On 09/20/04 , a Dell CPi notebook computer, serial # VLQLW, was found abandoned
 - it has a wireless PCMCIA card and an external homemade 802.11b antennae.
- It is **suspected** that this computer was used for hacking purposes
- It **may be** tied to a hacking suspect, **Greg Schardt**.
 - Schardt also goes by the online nickname of “Mr. Evil”
- Some of his associates have **said** that he would park his vehicle within range of Wireless Access Points (like Starbucks and other T-Mobile Hotspots) where he would then intercept internet traffic, attempting to get credit card numbers, usernames & passwords.

Goal: is Greg Schardt a hacker?

- Find
 - Any hacking software if there are any
 - Usage of evidence software
 - Any data that might have been generated.
 - intercepted internet traffic, credit card numbers, usernames & passwords
- Attempt to tie the computer to the suspect.

Get DD images

Download DD images

```
kali@kali: ~/hacking_case
kali@kali: ~/hacking_case 76x21

(kali㉿kali)-[~]
$ mkdir hacking_case

(kali㉿kali)-[~]
$ cd hacking_case

(kali㉿kali)-[~/hacking_case]
$ wget -q https://www.cfreds.nist.gov/images/hacking-dd/SCHARDT.001

(kali㉿kali)-[~/hacking_case]
$ wget -q https://www.cfreds.nist.gov/images/hacking-dd/SCHARDT.002
```

repeat `wget` 8 times...

```
(kali㉿kali)-[~/hacking_case]
$ wget -q https://www.cfreds.nist.gov/images/hacking-dd/SCHARDT.008
```

1. What is the image hash? Does the acquisition and verification hash match?

Merge .00? to one dd image

Check MD5

```
kali@kali: ~/hacking_case 76x22
(kali@kali)-[~/hacking_case]
$ ls
SCHARDT.001  SCHARDT.003  SCHARDT.005  SCHARDT.007
SCHARDT.002  SCHARDT.004  SCHARDT.006  SCHARDT.008

(kali@kali)-[~/hacking_case]
$ cat SCHARDT.00* > SCHARDT.dd

(kali@kali)-[~/hacking_case]
$ md5sum SCHARDT*
28a9b613d6eefe8a0515ef0a675bdebd  SCHARDT.001
c7227e7eea82d218663257397679a7c4  SCHARDT.002
ebba35acd7b8aa85a5a7c13f3dd733d2  SCHARDT.003
669b6636dcb4783fd5509c4710856c59  SCHARDT.004
c46e5760e3821522ee81e675422025bb  SCHARDT.005
99511901da2dea772005b5d0d764e750  SCHARDT.006
99511901da2dea772005b5d0d764e750  SCHARDT.007
8194a79a5356df79883ae2dc7415929f  SCHARDT.008
aee4fcd9301c03b3b054623ca261959a  SCHARDT.dd ←
```

2. What operating system was used on the computer?

Show partitions (NTFS) and unallocated space using *mmfs*

```
kali@kali: ~/hacking_case 76x22
(kali@kali)-[~/hacking_case]
$ mmfs SCHARDT.dd
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors
```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	-----	0000000000	0000000062	0000000063	Unallocated
002:	000:000	0000000063	0009510479	0009510417	NTFS / exFAT (0x07)
003:	-----	0009510480	0009514259	0000003780	Unallocated

Extract software, which contains OS information saved in Win registry

```
kali@kali: ~/hacking_case 76x22

(kali㉿kali)-[~/hacking_case]
$ fls -rF -o 63 SCHARDT.dd | grep -i software
r/r 9895-128-4: Program Files/Anonymizer/Toolbar/Images/software-A.bmp
r/r 9896-128-4: Program Files/Anonymizer/Toolbar/Images/software-D.bmp
r/r 9897-128-4: Program Files/Anonymizer/Toolbar/Images/software-M.bmp
r/r 6375-128-5: WINDOWS/PCHEALTH/HELPCTR/System/sysinfo/sysSoftwareInfo.htm
r/r 6376-128-5: WINDOWS/PCHEALTH/HELPCTR/System/sysinfo/sysSoftwareInfo.js
r/r 9742-128-4: WINDOWS/repair/software
r/r 336-128-4:  WINDOWS/system32/config/software
r/r 466-128-5:  WINDOWS/system32/config/software.LOG
r/r 471-128-3:  WINDOWS/system32/config/software.sav

(kali㉿kali)-[~/hacking_case]
$ icat -o 63 SCHARDT.dd 336 > software

(kali㉿kali)-[~/hacking_case]
$ ls software -l
-rw-r--r-- 1 kali kali 8650752 Feb 20 12:28 software
```

Find regrip plugin

```
kali@kali: ~/hacking_case 76x22  
(kaliⓈkali)-[~/hacking_case]  
$ rip.pl -l | grep -i winver  
149. winver v.20200525 [Software]
```

Find winver

```
(kaliⓈkali)-[~/hacking_case]  
$ rip.pl -r software -p winver  
Launching winver v.20200525  
winver v.20200525  
(Software) Get Windows version & build info  
  
ProductName           Microsoft Windows XP  
BuildLab               2600.xpclient.010817-1148  
RegisteredOrganization N/A  
RegisteredOwner       Greg Schardt  
InstallDate            2004-08-19 22:48:27Z
```

3. When was the OS install date?

- See previous slide
- InstallDate 2004-08-19 22:48:27Z

4. What is the timezone settings?

Search for "system"

```
(kali㉿kali)-[~/hacking_case]
$ fls -rF -o 63 SCHARDT.dd | egrep -i config/system$
r/r 334-128-4:  WINDOWS/system32/config/system
```

\$ To the end of the line.

Extract "system"

```
(kali㉿kali)-[~/hacking_case]
$ icat -o 63 SCHARDT.dd 334 > system
```

Find timezone

```
(kali㉿kali)-[~/hacking_case]
$ rip.pl -r system -p timezone
```

```
Launching timezone v.20200518
timezone v.20200518
(System) Get TimeZoneInformation key contents

TimeZoneInformation key
ControlSet001\Control\TimeZoneInformation
LastWrite Time 2004-08-19 17:20:02Z
    DaylightName    -> Central Daylight Time
    StandardName    -> Central Standard Time
    Bias            -> 360 (6 hours)
    ActiveTimeBias  -> 300 (5 hours)
```

5. Who is the registered owner?

```
kali@kali: ~/hacking_case 76x22

(kaliⓀkali)-[~/hacking_case]
$ rip.pl -l | grep -i winver
149. winver v.20200525 [Software]

(kaliⓀkali)-[~/hacking_case]
$ rip.pl -r software -p winver
Launching winver v.20200525
winver v.20200525
(Software) Get Windows version & build info

ProductName           Microsoft Windows XP
BuildLab               2600.xpclient.010817-1148
RegisteredOrganization N/A
RegisteredOwner       Greg Schardt
InstallDate            2004-08-19 22:48:27Z
```

6. What is the computer account name?

```
kali@kali: ~/hacking_case 76x26

(kali㉿kali)-[~/hacking_case]
$ rip.pl -r system -p compname
Launching compname v.20090727
compname v.20090727
(System) Gets ComputerName and Hostname values from System hive

ComputerName      = N-1A90DN6ZXK4LQ ←
TCP/IP Hostname   = n-1a9odn6zxk4lq
```

7. What is the primary domain (workgroup) name?

- Windows domain is associated with workgroup
 - and PC name, IP address
- Workgroup can be found in system event log
- We need to
 - search workgroup in system event log
 - extract the event log
 - find a tool to parse the event log
 - find domain information

Dump system event log

```
kali@kali: ~/hacking_case 76x28

(kali㉿kali)-[~/hacking_case]
$ fls -rF -o 63 SCHARDT.dd | egrep -i config/sysevent

r/r 3678-128-1: WINDOWS/system32/config/SysEvent.Evt

(kali㉿kali)-[~/hacking_case]
$ icat -o 63 SCHARDT.dd 3678 > SysEvent.Evt

(kali㉿kali)-[~/hacking_case]
$ ls -l SysEvent.Evt
-rw-r--r-- 1 kali kali 65536 Feb 20 16:55 SysEvent.Evt
```

Download evtparse

```
kali@kali: ~/hacking_case 76x28
(kali@kali)-[~/hacking_case]
$ git clone https://github.com/keydet89/Tools.git

Cloning into 'Tools'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 178 (delta 4), reused 6 (delta 4), pack-reused 166
Receiving objects: 100% (178/178), 8.08 MiB | 5.08 MiB/s, done.
Resolving deltas: 100% (99/99), done.
```

Test evtparse

```
(kali@kali)-[~/hacking_case]
$ perl Tools/source/evtparse.pl
evtparse [option]
Parse Event log (Win2000, XP, 2003)

-e file.....Event log (full path)
-d dir.....Directory where .evt files are located
-s .....Output in sequential format (record number and time
        generated values ONLY - use to see if system time may
        have been tampered with)
-t .....TLN output (default .csv)
-h .....Help (print this information)

Ex: C:\>evtparse -e secevent.evt -t > timeline.txt
    C:\>evtparse -e sysevent.evt -s
```

Parse System Event log

```
kali@kali: ~/hacking_case 76x28

(kali㉿kali)-[~/hacking_case]
$ perl Tools/source/evtparse.pl -e SysEvent.Evt -t > SysEvent.txt

(kali㉿kali)-[~/hacking_case]
$ ls -l SysEvent.*
-rw-r--r-- 1 kali kali 65536 Feb 20 16:55 SysEvent.Evt
-rw-r--r-- 1 kali kali 14183 Feb 20 17:20 SysEvent.txt
```

find domain information, PC name, etc.

kali@kali: ~/hacking_case 112x34

```
(kali@kali)-[~/hacking_case]
```

```
$ cat SysEvent.txt
```

1092934732	EVT	MACHINENAME	N/A	Serial/2;Info;\Device\Serial0,\Device\Serial0
1092934755	EVT	MACHINENAME	N/A	EventLog/6009;Info;5.01.,2600,,Uniprocessor Free
1092934755	EVT	MACHINENAME	N/A	EventLog/6005;Info;
1092935246	EVT	MACHINENAME	N/A	Serial/2;Info;\Device\Serial1,\Device\Serial1
1092954012	EVT	N-1A9ODN6ZXK4LQ	N/A	EventLog/6011;Info;MACHINENAME,N-1A9ODN6ZXK4LQ
1092954111	EVT	N-1A9ODN6ZXK4LQ	N/A	Dhcp/1007;Warn;0010A4933E09,169.254.242.213
1092954197	EVT	N-1A9ODN6ZXK4LQ	N/A	Workstation/3260;Info;workgroup,EVIL
1092955778	EVT	N-1A9ODN6ZXK4LQ	N/A	Setup/60054;Info;2600

8. When was the last recorded computer shutdown date/time?

```
kali@kali: ~/hac
(kaliⓈkali)-[~/hacking_case]
$ rip.pl -r system -p shutdown
Launching shutdown v.20200518
shutdown v.20200518
(System) Gets ShutdownTime value from System hive

ControlSet001\Control\Windows key, ShutdownTime value
LastWrite time: 2004-08-27 15:46:33Z
ShutdownTime : 2004-08-27 15:46:33Z
```

9. How many accounts are recorded (total number)?

- All Windows user account names, SIDs (Security Identifiers), login counts, creation dates, last password change dates, groups, and much more can be found in the Windows Registry SAM (Security Account Manager) file.

```
kali@kali:~$ (kaliⓈkali)-[~/hacking_case]
└─$ fls -rF -o 63 SCHARDT.dd | egrep -i config/sam

r/r 3667-128-4: WINDOWS/system32/config/SAM
r/r 3668-128-4: WINDOWS/system32/config/SAM.LOG

(kaliⓈkali)-[~/hacking_case]
└─$ icat -o 63 SCHARDT.dd 3667 > SAM
```

```
kali@kali: ~/hacking_case
(kali@kali)-[~/hacking_case]
$ rip.pl -r SAM -p samparse
Launching samparse v.20200825
samparse v.20200825
(SAM) Parse SAM file for user & group mbrshp info

User Information
-----
Username      : Administrator [500]
Full Name     :
User Comment  : Built-in account for administering the computer/domain
Account Type  : Default Admin User
Account Created : 2004-08-19 16:59:24Z
Name         :
Last Login Date : Never
Pwd Reset Date : 2004-08-19 17:17:29Z
Pwd Fail Date  : Never
Login Count   : 0
Embedded RID   : 500
--> Password does not expire
--> Normal user account

Username      : Guest [501]
Full Name     :
User Comment  : Built-in account for guest access to the computer/domain
Account Type  : Default Guest Acct
Account Created : 2004-08-19 16:59:24Z
Name         :
Last Login Date : Never
Pwd Reset Date : Never
Pwd Fail Date  : Never
Login Count   : 0
Embedded RID   : 501
--> Password does not expire
--> Password not required
--> Normal user account
```

```
Username      : HelpAssistant [1000]
Full Name     : Remote Desktop Help Assistant Account
User Comment  : Account for Providing Remote Assistance
Account Type  : Custom Limited Acct
Account Created : 2004-08-19 22:28:24Z
Name         :
Last Login Date : Never
Pwd Reset Date : 2004-08-19 22:28:24Z
Pwd Fail Date  : Never
Login Count   : 0
Embedded RID   : 1000
--> Password does not expire
--> Normal user account

Username      : SUPPORT_388945a0 [1002]
Full Name     : CN=Microsoft Corporation,L=Redmond,S=Washington,C=US
User Comment  : This is a vendor's account for the Help and Support Service
Account Type  : Custom Limited Acct
Account Created : 2004-08-19 22:35:19Z
Name         :
Last Login Date : Never
Pwd Reset Date : 2004-08-19 22:35:19Z
Pwd Fail Date  : Never
Login Count   : 0
Embedded RID   : 1002
--> Account Disabled
--> Password does not expire
--> Normal user account

Username      : Mr. Evil [1003]
Full Name     :
User Comment  :
Account Type  : Default Admin User
Account Created : 2004-08-19 23:03:54Z
Name         :
Last Login Date : 2004-08-27 15:08:23Z
Pwd Reset Date : 2004-08-19 23:03:54Z
Pwd Fail Date  : Never
Login Count   : 15
Embedded RID   : 1003
--> Password does not expire
--> Normal user account
```

10. What is the account name of the user who mostly uses the computer?

- Same as question 9

```
Username      : Mr. Evil [1003]
Full Name     :
User Comment  :
Account Type  : Default Admin User
Account Created : 2004-08-19 23:03:54Z
Name          :
Last Login Date : 2004-08-27 15:08:23Z
Pwd Reset Date : 2004-08-19 23:03:54Z
Pwd Fail Date  : Never
Login Count    : 15
Embedded RID   : 1003
--> Password does not expire
--> Normal user account
```

11. Who was
the last user
to logon to
the
computer?

Method 1

```
(kali㉿kali)-[~/hacking_case]
$ rip.pl -r software -p profilelist
Launching profilelist v.20200518
profilelist v.20200518
(Software) Get content of ProfileList key

Microsoft\Windows NT\CurrentVersion\ProfileList

Path      : %systemroot%\system32\config\systemprofile
SID       : S-1-5-18
LastWrite : 2004-08-19 22:48:26Z

Path      : %SystemDrive%\Documents and Settings\LocalService
SID       : S-1-5-19
LastWrite : 2004-08-27 15:08:21Z

Path      : %SystemDrive%\Documents and Settings\NetworkService
SID       : S-1-5-20
LastWrite : 2004-08-27 15:08:20Z

Path      : %SystemDrive%\Documents and Settings\Mr. Evil
SID       : S-1-5-21-2000478354-688789844-1708537768-1003
LastWrite : 2004-08-27 15:46:23Z
```

shutdown

Method 2

```
kali@kali: ~  
└─(kali@kali)-[~/hacking_case]  
└─$ rip.pl -r SAM -p samparse_tln _tln: timeline  
Launching samparse_tln v.20200826  
1092934764|SAM|Administrator|Acct Created (Default Admin User)  
1092935849|SAM|Administrator|Password Reset Date  
1092934764|SAM|Guest|Acct Created (Default Guest Acct)  
1092954504|SAM|HelpAssistant|Acct Created (Custom Limited Acct)  
1092954504|SAM|HelpAssistant|Password Reset Date  
1092954919|SAM|SUPPORT_388945a0|Acct Created (Custom Limited Acct)  
1092954919|SAM|SUPPORT_388945a0|Password Reset Date  
1092956634|SAM|Mr. Evil|Acct Created (Default Admin User)  
1092956634|SAM|Mr. Evil|Password Reset Date  
1093619303|SAM|Mr. Evil|Last Login (15)
```

epoch to a date

```
└─(kali@kali)-[~/hacking_case]  
└─$ date -d @1093619303  
Fri 27 Aug 2004 11:08:23 AM EDT
```

12. Proves that Greg Schardt is Mr. Evil

- A search (on local PC) for the name of “Greg Schardt” reveals multiple hits. One of these proves that Greg Schardt is Mr. Evil and is also the administrator of this computer. What file is it? What software program does this file relate to?
- Approach
 - Search “Greg” and check if the name associated with “Evil”
 - In order to search, we need to mount the DD image

Create a mounting point

Setup a loop device

Mount dd to the mounting point

```
(kali㉿kali)-[~/hacking_case]
$ sudo mkdir /mnt/loop
(kali㉿kali)-[~/hacking_case]
$ sudo losetup --partscan --find --show --read-only SCHARDT.dd
/dev/loop0

(kali㉿kali)-[~/hacking_case]
$ ls -l /dev/loop0*
brw-rw---- 1 root disk  7, 0 Feb 20 20:34 /dev/loop0
brw-rw---- 1 root disk 259, 0 Feb 20 20:34 /dev/loop0p1

(kali㉿kali)-[~/hacking_case]
$ mount /dev/loop0p1 /mnt/loop
mount: /mnt/loop: must be superuser to use mount. ✗ missing sudo

(kali㉿kali)-[~/hacking_case]
$ sudo mount /dev/loop0p1 /mnt/loop
Error opening '/dev/loop0p1' read-write ✓ error message, but works
Could not mount read-write, trying read-only

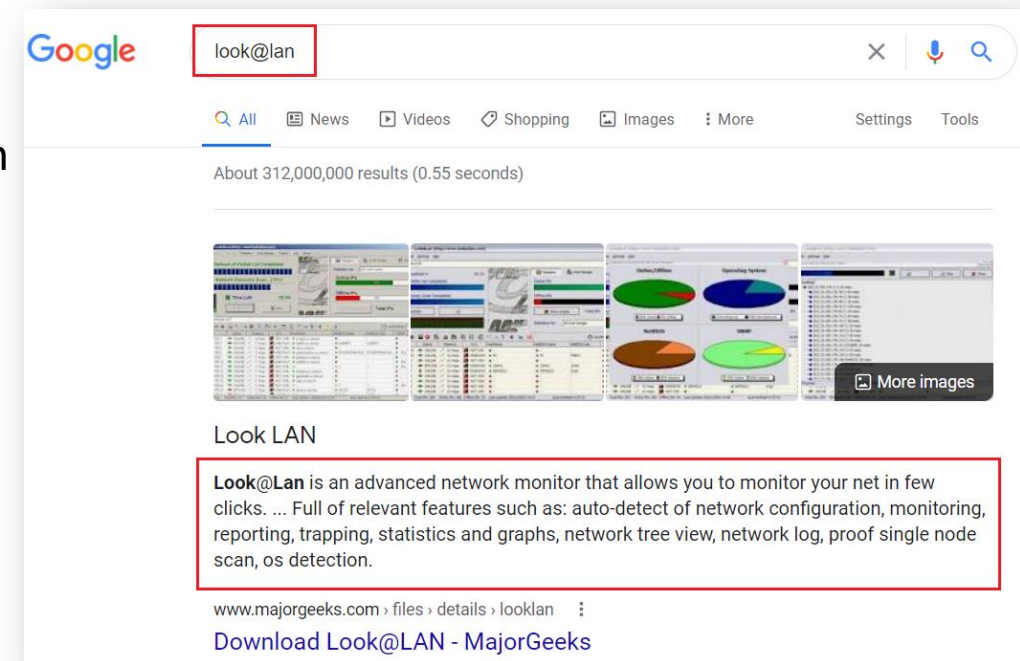
(kali㉿kali)-[~/hacking_case]
$ ls /mnt/loop
AUTOEXEC.BAT  BOOTSECT.DOS  'Documents and Settings'  MSDOS.---
boot.ini      COMMAND.COM   FRUNLOG.TXT               MSDOS.SYS
BOOTLOG.PRIV  CONFIG.SYS    hiberfil.sys              'My Documents'
BOOTLOG.TXT   DETLOG.TXT    IO.SYS                    NETLOG.TXT
```

Search globally for the string "Greg Schardt"

```
kali@kali: ~  
└─(kali@kali)-[~/hacking_case]  
└─$ grep -rn '/mnt/loop/' -e 'Greg Schardt' -r: recursive -n: line number  
/mnt/loop/Program Files/Look@LAN/irunin.ini:29:%REGOWNER%=Greg Schardt  
/mnt/loop/Program Files/Look@LAN/irunin.ini:396:%USERNAME%=Greg Schardt  
/mnt/loop/WINDOWS/Look@LAN Setup Log.txt:42:Value data = Greg Schardt
```

Observation:

- look@lan is a software because it was installed under the "Program Files" folder
- look@lan has a configuration file named "irunin.ini"
- The initial config file has owner and user name information.
- The config file has a setup log



Search for the string "evil"

kali@kali: ~/haci

```
(kali@kali)-[~/hacking_case]
```

```
$ cat "/mnt/loop/Program Files/Look@LAN/irunin.ini" | grep -i 'evil'
```

```
%LANUSER%=Mr. Evil
```

```
%DESKTOP%=C:\Documents and Settings\Mr. Evil\Desktop
```

```
%STARTMENU%=C:\Documents and Settings\Mr. Evil\Start Menu
```

```
%STARTMENUPROGRAMS%=C:\Documents and Settings\Mr. Evil\Start Menu\Programs
```

```
%STARTUP%=C:\Documents and Settings\Mr. Evil\Start Menu\Programs\Startup
```

```
%MYDOCUMENTSDIR%=C:\Documents and Settings\Mr. Evil\My Documents
```

```
%SRCFILE%=C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe
```

```
%SRCDIR%=C:\Documents and Settings\Mr. Evil\Desktop
```

```
(kali@kali)-[~/hacking_case]
```

```
$ cat "/mnt/loop/WINDOWS/Look@LAN Setup Log.txt" | grep -i 'evil'
```

```
C:\Documents and Settings\Mr. Evil\Desktop\Look@LAN.lnk
```

```
C:\Documents and Settings\Mr. Evil\Desktop\Look@Host.lnk
```

Other evidence may helpful:

Search both key words ""Evil"" and ""Greg"". The search results show Evil is associated with yahoo web visits.

```
(kali㉿kali)-[~/hacking_case]
$ grep -rni '/mnt/loop/' -e '\bEvil\b.*\bGreg\b' \b xxxx \b: word boundry
Binary file /mnt/loop/Documents and Settings/Mr. Evil/Local Settings/History/History.IE5/index.dat matches
Binary file /mnt/loop/Documents and Settings/Mr. Evil/Local Settings/History/History.IE5/MSHist012004081620040823/index.dat matches

(kali㉿kali)-[~/hacking_case]
$ strings '/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/History/History.IE5/index.dat' | grep -i greg
Visited: Mr. Evil@http://edit.yahoo.com/config/id_check?.fn=Greg&.ln=Schardt&.id=mrevil2000&.u=b568cfp0ic6g0

(kali㉿kali)-[~/hacking_case]
$ strings '/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/History/History.IE5/MSHist012004081620040823/index.dat' | grep -i 'greg'
:2004081620040823: Mr. Evil@http://edit.yahoo.com/config/id_check?.fn=Greg&.ln=Schardt&.id=mrevil2000&.u=b568cfp0ic6g0
```

13. List the network cards used by this computer

```
kali@kali: ~/hacking_case 149
(kaliⓈkali)-[~/hacking_case]
$ rip.pl -r software -p networkcards

Launching networkcards v.20200518
networkcards v.20200518
(Software) Get NetworkCards Info

NetworkCards
Microsoft\Windows NT\CurrentVersion\NetworkCards

Description                                     Key LastWrite time
Compaq WL110 Wireless LAN PC Card                2004-08-27 15:31:44Z
Xircom CardBus Ethernet 100 + Modem 56 (Ethernet Interface) 2004-08-19 17:07:19Z
```

14. This same file reports the IP address and MAC address of the computer. What are they?

- list all files contain IP address
- list all files contain MAC address
- Find intersection of two files

egrep or **grep -E**: Run grep with extended regular expressions.
-r: Recursively search
I: ignore binary file
l: only list file name

```
kali@kali: ~/hacking_case 112x36

(kali㉿kali)-[~/hacking_case]
$ egrep -rIl '\b[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\b' '/mnt/loop/' > ip.txt

(kali㉿kali)-[~/hacking_case]
$ egrep -rIl '[^-/]\b[0-9a-fA-F]{12}\b' '/mnt/loop' > mac.txt

(kali㉿kali)-[~/hacking_case]
$ comm -12 ip.txt mac.txt
comm: file 1 is not in sorted order
/mnt/loop/Program Files/Look@LAN/irunin.ini
/mnt/loop/Program Files/mIRC/channels/channels.txt
comm: file 2 is not in sorted order
comm: input is not in sorted order
```

\b: Match either the beginning or end of a word.
[^]: match anything except the characters within a bracket by beginning the list of characters within the brackets with a **^** character.

Verify irunin.ini has both IP and MAC addresses

```
kali@kali: ~/hacking_case 85x36

(kali㉿kali)-[~/hacking_case]
$ egrep -r '\b[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\b' '/mnt/loop/Program Files/Look@LAN/irunin.ini'
%LANIP%=192.168.1.111

(kali㉿kali)-[~/hacking_case]
$ egrep -r '^[^-/]\b[0-9a-fA-F]{12}\b' '/mnt/loop/Program Files/Look@LAN/irunin.ini'
%LANNIC%=0010a4933e09
```

15. Which NIC card was used during the installation and set-up for LOOK@LAN?

- MAC addresses are primarily assigned by device manufacturers
 - Often referred to as the burned-in address.
- The IEEE assigns manufacturers the numbers, called organizationally unique identifiers (6 Hex).
 - The remaining bits of each MAC address act as a serial number, assigned to a NIC when it is made.

https://www.adminsub.net/mac-address-finder/XIRCOM

adminsub.net/mac-address-finder/XIRCOM

adminsubnet

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IPv4 Subnet Calculator Password Generator/Decryptor MAC

MAC Address Finder

MAC address or vendor:

Enter **first 6 characters** or **full** MAC address. Or search by Vendor name, e.g. **cisco** or **apple**

Database updated - April 25, 2020

Search results for "XIRCOM"

MAC	Vendor
0080C7	XIRCOM
0010A4	XIRCOM
00053C	XIRCOM

%LANNIC%=0010a4933e09

It proves that the XIRCOM was used during the installation and set-up for LOOK@LAN.

16. Find 6 installed programs that may be used for hacking.

```
kali@kali: ~/hacking

(kali@kali)-[~/hacking_case]
$ rip.pl -r software -p uninstall

Launching uninstall v.20200525
uninstall v.20200525
(Software, NTUSER.DAT) Gets contents of Uninstall

Uninstall
Microsoft\Windows\CurrentVersion\Uninstall

2004-08-27 15:29:19Z
Ethereal 0.10.6 v.0.10.6

2004-08-27 15:15:19Z
WinPcap 3.01 alpha

2004-08-27 15:12:15Z
Network Stumbler 0.4.0 (remove only)

2004-08-25 15:56:11Z
Look@LAN 2.50 Build 29

2004-08-20 15:13:08Z
123 Write All Stored Passwords

2004-08-20 15:12:43Z
Powertoys For Windows XP v.1.00.0000
```

```
2004-08-20 15:09:02Z
CuteFTP

2004-08-20 15:08:19Z
Forté Agent

2004-08-20 15:07:25Z
Faber Toys v.2.4 Build 216

2004-08-20 15:05:58Z
Cain & Abel v2.5 beta45

2004-08-20 15:05:09Z
Anonymizer Bar 2.0 (remove only)

2004-08-19 23:04:50Z
WebFldrs XP v.9.50.5318
```

Hacking Programs Explained

- Ethereal: “Ethereal is a sniffer, an application that can capture all packets sent through a network to then interpret them and see what tasks are being carried out on the network.”
- Network Stumbler: “Network stumbler is a wifi scanner and monitor tool for windows, it allows you to detect WLANs using 802.11a, 802.11b and 802.11g wireless cards.”
- WinPcap: “Enables applications to send and receive raw network packets to/from network cards. Receiving raw network packets is also known as packet capturing, therefore the name "Windows Packet capture library".
- 123 Write All Stored: Password Cracker. “A Password Cracker program is used to decrypt a password or password file. Some Password Crackers programmatically try different passwords while others use brute force logging in with passwords word lists.”
- Anonymizer Bar 2.0: “The Anonymizer toolbar plug-in hosts seven tools that basically switch your originating Web address to the Anonymizer proxy server, thus concealing your identity.”
- Cain and Abel: “The Cain & Abel program is designed to crack passwords in the most difficult situations. Capable of detecting passwords hidden by "asterisks. [Gives user] access to a sniffer - a network traffic analyzer, the purpose of which is to intercept valuable information transmitted over the network.”
- CuteFTP: “Allows you to transfer files from your PC to another PC or a server in an easy way, just drag&drop and see how they are transferred.”

17. What is the SMTP email address for Mr. Evil?

- **NTUSER.DAT** Contains Your **User Profile Settings**
- Every time you make a change to the look and behavior of Windows and installed programs, Windows needs to remember your preferences the next time it loads.
 - whether that's your desktop background, monitor resolution, or even which printer is the default,
- Windows accomplishes this by first storing that information to the **HKEY_CURRENT_USER** hive.
 - Then when you sign out or shut down, Windows saves that information to the NTUSER.DAT file.
 - The next time you sign in, Windows will load NTUSER.DAT to memory, and all your preferences load to the Registry again.

Search for ntuser.data

```
kali@kali: ~/hacking_case 92x40
(kali@kali)-[~/hacking_case]
$ fls -rF -o 63 SCHARDT.dd | grep -i 'ntuser.dat'

r/r 7324-128-4: Documents and Settings/Default User/NTUSER.DAT
r/r 391-128-4: Documents and Settings/LocalService/NTUSER.DAT
r/r 418-128-4: Documents and Settings/LocalService/ntuser.dat.LOG
r/r 345-128-4: Documents and Settings/Mr. Evil/NTUSER.DAT ←
r/r 9798-128-4: Documents and Settings/Mr. Evil/ntuser.dat.LOG
r/r 350-128-4: Documents and Settings/NetworkService/NTUSER.DAT
r/r 377-128-4: Documents and Settings/NetworkService/ntuser.dat.LOG
r/r 9746-128-4: WINDOWS/repair/ntuser.dat
```

Extract Evil ntuser.data

```
(kali@kali)-[~/hacking_case]
$ icat -o 63 SCHARDT.dd 345 > NTUSER_Evil.DAT
```

Search the email pattern

```
(kali@kali)-[~/hacking_case]
$ strings NTUSER_Evil.DAT | grep -iP '\b^[\w\.-]+@([\w-]+\.)+[\w-]{2,4}\b'
whoknowsme@sbcglobal.net
xe@shdoclc.dll,-866
Look@LAN.lnk
Look@LAN.lnk
Look@LAN.lnk
```

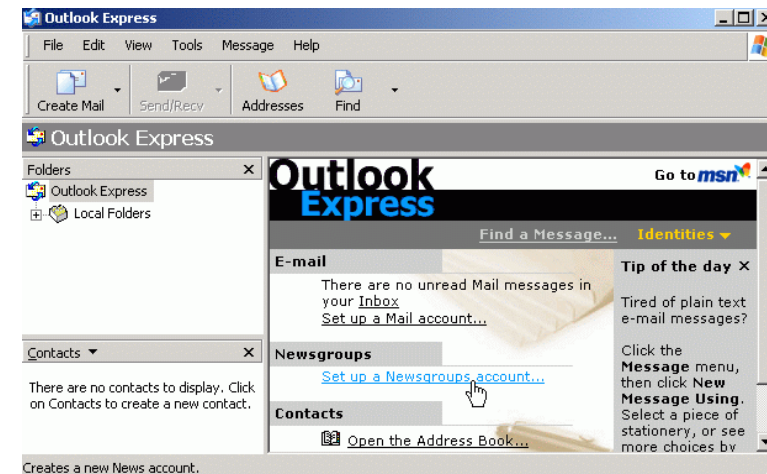
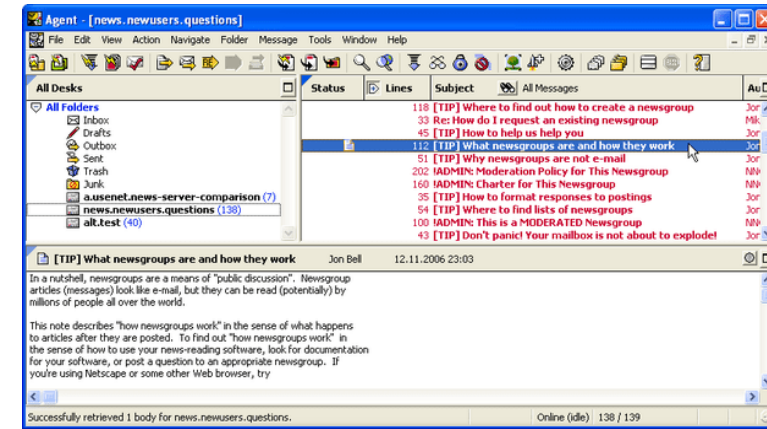
The **strings** command returns each string of printable characters in files. Its main uses are to determine the contents of and to extract text from binary files.

18. What are the NNTP (news server) settings for Mr. Evil?

- Understand new server
- Find new applications installed
- Find the application installation directory
- Search for configuration files or key words

Understand new server

- The Network News Transfer Protocol (NNTP) is an application protocol
 - for transporting Usenet news articles (netnews) between news servers and
 - for reading and posting articles by end user client applications.
- Examples
 - Commercial: NewsLeecher, Forte Agent, Lotus Notes, BinTube
 - Free: Pan, Spotnet, Claws Mail, Gnus, Outlook Express



Find new applications installed

```
kali@kali: ~/hacking_case 92x40
(kaliⓈkali)-[~/hacking_case]
$ rip.pl -r software -p uninstall

Launching uninstall v.20200525
uninstall v.20200525
(Software, NTUSER.DAT) Gets contents of Uninstall keys

2004-08-20 15:08:19Z
  Forté Agent ←
2004-08-19 22:31:51Z
  AddressBook
  ICW
  OutlookExpress ←
```

Search for installed **Forte Agent** using the keyword “forte” or “agents”

kali@kali: ~/hacking_case 92x40

```
(kali㉿kali)-[~/hacking_case]
```

```
$ fls -rF -o 63 SCHARDT.dd | grep -i 'forte'
```

nothing

```
(kali㉿kali)-[~/hacking_case]
```

```
$ fls -rF -o 63 SCHARDT.dd | grep -i 'agent' | head
```

```
r/r 10064-128-1: Documents and Settings/Mr. Evil/Desktop/Tools/Agent.lnk
r/r 10065-128-4: Documents and Settings/Mr. Evil/Start Menu/Programs/Agent
/Agent Help.lnk
r/r 10066-128-1: Documents and Settings/Mr. Evil/Start Menu/Programs/Agent
/Readme.lnk
r/r 10210-128-3: My Documents/ARCHIVE/Arj/AGENTS.TXT
r/r 10055-128-3: Program Files/Agent/8859-1.cod
r/r 10057-128-3: Program Files/Agent/8859-15.cod
r/r 10056-128-3: Program Files/Agent/8859-1w.cod
r/r 10013-128-3: Program Files/Agent/agent.cnt
r/r 10009-128-3: Program Files/Agent/agent.exe ←
r/r 10012-128-3: Program Files/Agent/agent.hlp
```

Looking for **Forte Agent** configuration files or data

```
kali@kali: ~/hacking_case 92x40
(kali@kali)-[~/hacking_case]
$ fls -rF -o 63 SCHARDT.dd | grep -i 'Program Files/Agent/'
r/r 10055-128-3: Program Files/Agent/8859-1.cod
r/r 10057-128-3: Program Files/Agent/8859-15.cod
r/r 10056-128-3: Program Files/Agent/8859-1w.cod
r/r 10013-128-3: Program Files/Agent/agent.cnt
r/r 10009-128-3: Program Files/Agent/agent.exe
r/r 10012-128-3: Program Files/Agent/agent.hlp
r/r 10016-128-3: Program Files/Agent/All_COD/8859-1.cod
r/r 10017-128-3: Program Files/Agent/All_COD/8859-10.cod
r/r 11406-128-4: Program Files/Agent/Data/AGENT.INI ←
r/r 11416-128-1: Program Files/Agent/Data/errorlog.txt
r/r 11420-128-1: Program Files/Agent/Data/FILTERS.DAT
r/r 11423-128-1: Program Files/Agent/Data/FILTERS.IDX
r/r 11727-128-3: Program Files/Agent/Data/GROUPS.DAT
r/r 11728-128-3: Program Files/Agent/Data/GROUPS.IDX
r/r 11417-128-3: Program Files/Agent/Data/GRPDAT.BAK
```

Find news server configuration

```
kali@kali: ~/hacking_case 92x38
(kali@kali)-[~/hacking_case]
$ icat -o 63 SCHARDT.dd 11406 | more
;AGENT.INI
;
;For information about the settings in this file,
;search for AGENT.INI in the online help.

[Profile]
Build="32.560"
FullName="Mr Evil"
EmailAddress="whoknowsme@sbcglobal.net" ←
EmailAddressFormat=0
ReplyTo=""
Organization="N/A"
DoAuthorization=1
SavePassword=1
UserName="whoknowsme@sbcglobal.net"
Password="84106D94696F"
SMTPLoginProtocol=2
SMTPUsePOPLogin=0
SMTPUserName="whoknowsme@sbcglobal.net"
SMTPSavePassword=1
SMTPPassword="84106D94696F"
```

```
SMTPPassword="84106D94696F"
IsRegistered=0
IsRegistered19=0
IsLicensed=3
Key=""
EnableSupportMenu=0

[Servers]
NewsServer="news.dallas.sbcglobal.net" ←
MailServer="smtp.sbcglobal.net" ←
POPServer=""
NNTPPort=119
SMTPPort=25
POPPort=110
SMTPServerPort=25
```

List all files that are associated with **outlook express** by searching the key work “outlook”

kali@kali: ~/hacking_case 92x38

```
(kali@kali)-[~/hacking_case]
```

```
$ find -rF -o 63 SCHARDT.dd | grep -i 'outlook'
```

2 ⚙

```
r/r 11431-128-3: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/cleanup.log
r/r 11443-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.cardz.dbx
r/r 11444-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.codez.dbx
r/r 11445-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.crackz.dbx
r/r 11442-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.dbx
r/r 11539-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.hackerz.dbx
r/r 11446-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.moderated.dbx
r/r 11536-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.phreakz.dbx
r/r 11523-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.2600.programz.dbx
r/r 11505-128-4: Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/alt.binaries.hacking.beginner.dbx
```

Find the news server: **news.dallas.sbcglobal.net** in the .dbx (outlook express format)

[illegible]

19. What two installed programs show this information?

```
kali@kali: ~/hacking_case 92x40
(kaliⓈkali)-[~/hacking_case]
$ rip.pl -r software -p uninstall

Launching uninstall v.20200525
uninstall v.20200525
(Software, NTUSER.DAT) Gets contents of Uninstall keys

2004-08-20 15:08:19Z
  Forté Agent ←
2004-08-19 22:31:51Z
  AddressBook
  ICW
  OutlookExpress ←
```

20. List 5 newsgroups that Mr. Evil has subscribed to?

```
kali@kali: ~/hacking_case 92x33
(kaliⓈkali)-[~/hacking_case]
$ ls '/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Application Data/Identities/{EF086998-1115-4ECD-9B13-9ADC067B4929}/Microsoft/Outlook Express/'
alt.2600.cardz.dbx
alt.2600.codez.dbx
alt.2600.crackz.dbx
alt.2600.dbx
alt.2600.hackerz.dbx
alt.2600.moderated.dbx
alt.2600.phreakz.dbx
alt.2600.programz.dbx
alt.binaries.hacking.beginner.dbx
alt.binaries.hacking.computers.dbx
alt.binaries.hacking.utilities.dbx
alt.binaries.hacking.websites.dbx
alt.dss.hack.dbx
alt.hacking.dbx
alt.nl.binaries.hack.dbx
alt.stupidity.hackers.malicious.dbx
cleanup.log
'Deleted Items.dbx'
Folders.dbx
free.binaries.hackers.malicious.dbx
free.binaries.hacking.beginner.dbx
free.binaries.hacking.computers.dbx
free.binaries.hacking.talentless.troll-haven.dbx
free.binaries.hacking.talentless.troll_haven.dbx
free.binaries.hacking.utilities.dbx
free.binaries.hacking.websites.dbx
Inbox.dbx
Offline.dbx
Outbox.dbx
```

21. Investigate an Internet Relay Chat program

A popular IRC (Internet Relay Chat) program called MIRC was installed. What are the user settings that was shown when the user was online and in a chat channel?

- Find new applications installed
- Find the application installation directory
- Search for configuration files or key words

Find mIRC installation directory

```
(kali㉿kali)-[~/hacking_case]
$ fls -rF -o 63 SCHARDT.dd | grep -i 'mirc'
r/r 10087-128-4: Documents and Settings/All Users/Start Menu/Programs/mIRC/IRC Intro.lnk
r/r 10086-128-4: Documents and Settings/All Users/Start Menu/Programs/mIRC/mIRC Help.lnk
r/r 10088-128-4: Documents and Settings/All Users/Start Menu/Programs/mIRC/Readme.txt.lnk
r/r 10089-128-4: Documents and Settings/All Users/Start Menu/Programs/mIRC/Versions.txt.lnk
r/r 10085-128-1: Documents and Settings/Mr. Evil/Desktop/Tools/mIRC.lnk
r/r 10081-128-1: Program Files/mIRC/aliases.ini
r/r 11072-128-6: Program Files/mIRC/channels/channels.txt
r/r 10077-128-3: Program Files/mIRC/ircintro.hlp
r/r 11315-128-4: Program Files/mIRC/logs/#Chataholics.UnderNet.log
r/r 11316-128-4: Program Files/mIRC/logs/#CyberCafe.UnderNet.log
r/r 11276-128-1: Program Files/mIRC/logs/#Elite.Hackers.UnderNet.log
r/r 11074-128-1: Program Files/mIRC/logs/#evilfork.EFnet.log
r/r 11401-128-1: Program Files/mIRC/logs/#funny.UnderNet.log
r/r 11306-128-1: Program Files/mIRC/logs/#houston.UnderNet.log
r/r 11073-128-1: Program Files/mIRC/logs/#ISO-WAREZ.EFnet.log
r/r 11272-128-4: Program Files/mIRC/logs/#LuxShell.UnderNet.log
r/r 11273-128-4: Program Files/mIRC/logs/#mp3xserv.UnderNet.log
r/r 11327-128-4: Program Files/mIRC/logs/#thedarktower.AfterNET.log
r/r 11275-128-1: Program Files/mIRC/logs/#ushells.UnderNet.log
r/r 11317-128-1: Program Files/mIRC/logs/m5tar.UnderNet.log
r/r 10074-128-6: Program Files/mIRC/mirc.exe
r/r 10073-128-3: Program Files/mIRC/mirc.hlp
r/r 10080-128-3: Program Files/mIRC/mirc.ini
r/r 10082-128-3: Program Files/mIRC/popups.ini
r/r 10078-128-3: Program Files/mIRC/readme.txt
r/r 10083-128-3: Program Files/mIRC/servers.ini
r/r 10084-128-5: Program Files/mIRC/urls.ini
r/r 10079-128-3: Program Files/mIRC/versions.txt
r/r 11071-128-4: WINDOWS/Prefetch/MIRC.EXE-0661EC22.pf
r/r 10090-128-4: WINDOWS/Prefetch/MIRC612.EXE-02791C37.pf
```

Find some important configuration

```
(kali㉿kali)-[~/hacking_case]
```

```
$ icat -o 63 SCHARDT.dd 10080 | grep -iE 'user|email|log|ip|server'
```

```
n46=#mIRCScripts
```

```
n75=#UserGuide,"The official Undernet help channel"
```

```
n76=#UserHelp
```

```
accept=*.bmp,*.gif,*.jpg,*.log,*.mid,*.mp3,*.png,*.txt,*.wav,*.wma,*.zip
```

```
logdir=logs\
```

```
userid=Mrevil
```

```
useip=yes
```

```
status=/lusers
```

```
ServerStatus=on
```

```
[fileserver]
```

```
[dccserver]
```

```
user=Mini Me
```

```
email=none@of.ya
```

```
host=Undernet: US, CA, LosAngelesSERVER:losangeles.ca.us.undernet.org:66600
```

```
servers=servers.ini
```

This IRC program has the capability to log chat sessions.

22. List 3 IRC channels that the user of this computer accessed.

```
kali@kali: ~/hacking_case 98x39  
  
(kali㉿kali)-[~/hacking_case]  
$ fls -rF -o 63 SCHARDT.dd | grep -i 'Program Files/mIRC/logs'  
r/r 11315-128-4: Program Files/mIRC/logs/#Chataholics.UnderNet.log  
r/r 11316-128-4: Program Files/mIRC/logs/#CyberCafe.UnderNet.log  
r/r 11276-128-1: Program Files/mIRC/logs/#Elite.Hackers.UnderNet.log  
r/r 11074-128-1: Program Files/mIRC/logs/#evilfork.EFnet.log  
r/r 11401-128-1: Program Files/mIRC/logs/#funny.UnderNet.log  
r/r 11306-128-1: Program Files/mIRC/logs/#houston.UnderNet.log  
r/r 11073-128-1: Program Files/mIRC/logs/#ISO-WAREZ.EFnet.log  
r/r 11272-128-4: Program Files/mIRC/logs/#LuxShell.UnderNet.log  
r/r 11273-128-4: Program Files/mIRC/logs/#mp3xserv.UnderNet.log  
r/r 11327-128-4: Program Files/mIRC/logs/#thedarktower.AfterNET.log  
r/r 11275-128-1: Program Files/mIRC/logs/#ushells.UnderNet.log  
r/r 11317-128-1: Program Files/mIRC/logs/m5tar.UnderNet.log
```

23. Investigate Ethereal

- Ethereal, a popular “sniffing” program that can be used to intercept wired and wireless internet packets was also found to be installed.
- When TCP packets are collected and re-assembled, the default save directory is that users **default documents** directory.
- What is the name of the file that contains the intercepted data?
 - Find the location **\Document and Setting\Mr. Evil**
 - Search for TCP packets saved (.pcap)

Find the type of file using **file** command

```
kali@kali: ~ 84x29

(kali㉿kali)-[~]
$ ls /mnt/loop/Documents\ and\ Settings/Mr.\ Evil
'Application Data'  Favorites  'My Documents'  ntuser.dat.LOG  Recent
Templates
Cookies            interception  NetHood         ntuser.ini      SendTo
Desktop            'Local Settings'  NTUSER.DAT      PrintHood       'Start Menu'
```



```
(kali㉿kali)-[~]
$ file /mnt/loop/Documents\ and\ Settings/Mr.\ Evil/interception
/mnt/loop/Documents and Settings/Mr. Evil/interception: pcap capture file, microseconds (little-endian) - version 2.4 (Ethernet, capture length 65535)
```



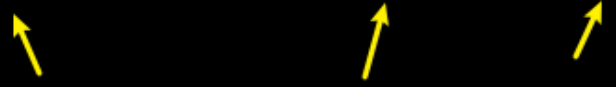
24. What type of device was the victim (person who had his internet surfing recorded) using?

```
(kali㉿kali)-[~]  
$ tshark --help  
TShark (Wireshark) 3.2.7 (Git v3.2.7 packaged as 3.2.7-1)  
Dump and analyze network traffic.  
See https://www.wireshark.org for more information.  
  
Usage: tshark [options] ...  
  
Input file:  
  -r <infile>, --read-file <infile> ←  
                                     set the filename to read from (or '-' for stdin)  
  -e <field> ←  
                                     field to print if -Tfields selected (e.g. tcp.port,  
                                     _ws.col.Info)  
                                     this option can be repeated to print multiple fields  
  -Y <display filter>, --display-filter <display filter>  
                                     packet display filter in Wireshark display filter  
                                     syntax
```

Search for user agent from http request

```
kali@kali: ~ 102x37

(kaliⓈkali)-[~]
$ tshark -r /mnt/loop/Documents\ and\ Settings/Mr.\ Evil/interception -Y http.request -T fields -e http.user_agent -e http.host
```



Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	www.passportimages.com
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)	mobile.msn.com

25. What websites was the victim accessing?

Sort the requests with unique entries

```
kali@kali: ~ 100x37
(kali@kali)-[~]
$ tshark -r /mnt/loop/Documents\ and\ Settings/Mr.\ Evil/interception -Y http.request -T fields
-e http.user_agent -e http.host | sort -u
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)
Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC; 240x320)
```

unique entries

login.passport.com
login.passport.net
mobile.msn.com
www.passportimages.com

26. Search for the main user's web-based email address. What is it?

Understand Extended Regular Expression (ERE)

```
(kali㉿kali)-[~]  
$ grep --help  
Usage: grep [OPTION]... PATTERNS [FILE]...  
Search for PATTERNS in each FILE.  
Example: grep -i 'hello world' menu.h main.c  
PATTERNS can contain multiple patterns separated by newlines.  
  
Pattern selection and interpretation:  
-E, --extended-regexp    PATTERNS are extended regular expressions  
-F, --fixed-strings      PATTERNS are strings  
-G, --basic-regexp       PATTERNS are basic regular expressions  
-P, --perl-regexp        PATTERNS are Perl regular expressions  
-e, --regexp=PATTERNS    use PATTERNS for matching  
-f, --file=FILE          take PATTERNS from FILE  
-i, --ignore-case        ignore case distinctions in patterns and data  
                        --no-ignore-case    do not ignore case distinctions (default)  
-o, --only-matching      show only nonempty parts of lines that match  
-r, --recursive          like --directories=recurse  
-h, --no-filename        suppress the file name prefix on output  
-I                        equivalent to --binary-files=without-match
```

1. Basic vs Extended Regular Expressions

- In basic regular expressions the meta-characters `?`, `+`, `{`, `|`, `(`, and `)` lose their special meaning; instead use the backslashed versions `\?`, `\+`, `\{`, `\|`, `\(`, and `\)`.

2. Pre-defined classes

- `[:alnum:]`, `[:alpha:]`, `[:blank:]`,
- `[:cntrl:]`, `[:digit:]`, `[:graph:]`,
- `[:lower:]`, `[:print:]`, `[:punct:]`,
- `[:space:]`, `[:upper:]`, `[:xdigit:]`

```

(kali㉿kali)-[~]
$ grep -EiorhI '([[:alnum:]]_.-]+@[[:alnum:]]_.-]+?\.([[:alpha:]]{2,6})'
'/mnt/loop/Documents and Settings/Mr. Evil/'
mrevilrulez@yahoo.com
mrevilrulez@yahoo.com
mrevilrulez@yahoo.com
mailbot@yahoo.com
mrevilrulez@yahoo.com
mrevilrulez@yahoo.com
mrevilrulez@yahoo.com
nightwolf@confine.com
mrevilrulez@yahoo.com
mrevilrulez@yahoo.com
mrevilrulez@yahoo.com
seabach@shaw.ca
J06.42522@pd7tw2no...
NOSPAM-fred@warddriving.com
jim@mcmahon.cc
jim@mcmahon.cc
hacked@2600.com
webmaster@2600.com
you@your-name.com
LmT@marijuana.com
chris@splitinfinity.com
NOSPAM-fred@warddriving.com
123@123.com
info@mosnews.com
info@mosnews.com
--Rating@Mail.ru
drudge@drudgereport.com
DRUDGE@DRUDGEREPORT.COM
_RATED_9.5_@_Warez.com

```

- E: Extended Regular Expression
- i: ignore case
- o: show only match, not whole lines
- r: recursive
- h: don't show file name
- l: ignore binary files

```

T50admin@usa.net
jim@mcmahon.cc
chillen@hoo.com
img4i0lqhsh6n7hlqth96lfd5jd1acjrh9@4ax.com...
beatnik@mail.gr
teandson@aol.com
9a64i0p9vk73bpmnq4s40iq6asem5k80er@4ax.com
corenode01a@yahoo.removethisfirst.com
mrevilrulez@yahoo.com
mrevilrulez@yahoo.com
cathomas@msn.com
Oi.11@fed1read04...
mauddib@dune.com
dqbug010mo29ufsbo4dq491vvihucqfh69@4ax.com...
heyjude18@hotmail.com...as
hp01@mailadded.com
logaritmo50@yahoo.com
logaritmo50@hotmail.com
PASSADMINBOT@HOTMAIL.COM
HERE@HOTMAIL.COM
PASSCODE@HOTMAIL.COM
slim532@hotmail.com
248e504e.0408150655.a30aac9@posting.google.com...
PASSADMINBOT@HOTMAIL.COM
HERE@HOTMAIL.COM
PASSCODE@HOTMAIL.COM
slim532@hotmail.com
248e504e.0408150655.a30aac9@posting.google.com...
PASSADMINBOT@HOTMAIL.COM
HERE@HOTMAIL.COM
PASSCODE@HOTMAIL.COM
suckme@oyea.lick
president@whitehouse.gov
tmt3i0tnq18gm819ecv27r73vm6hnoddcn@4ax.com...
suckme@oyea.lick
info@mosnews.com
info@mosnews.com
--Rating@Mail.ru
webmaster@2600.com
webmaster@2600.com

```

Sort all emails, count them, and sort again based on counts

```
kali@kali: ~ 73x46

(kali@kali)-[~]
$ grep -EiorhI '([[:alnum:]]_.-]+@[[:alnum:]]_.-]+?\.[[:alpha:]]{2,6})'
'/mnt/loop/Documents and Settings/Mr. Evil/' | sort | uniq -c | sort -nr
12 mrevilrulez@yahoo.com
6 info@mosnews.com
4 jim@mcmahon.cc
3 webmaster@2600.com
3 --Rating@Mail.ru
3 PASSCODE@HOTMAIL.COM
3 PASSADMINBOT@HOTMAIL.COM
3 NOSPAM-fred@wardriving.com
3 HERE@HOTMAIL.COM
2 suckme@oyea.lick
2 slim532@hotmail.com
2 248e504e.0408150655.a30aac9@posting.google.com...
1 you@your-name.com
1 tmt3i0tnq18gm819ecv27r73vm6hnoddcn@4ax.com...
1 tH1.10237466@twister.southeast.rr.com...
1 teandson@aol.com
1 T50admin@usa.net
1 seabach@shaw.ca
1 _RATED_9.5_@_Warez.com
```

uniq options:

-c, --count

prefix lines by the number of occurrences

sort options:

-n, --numeric-sort

compare according to string numerical value

-r, --reverse

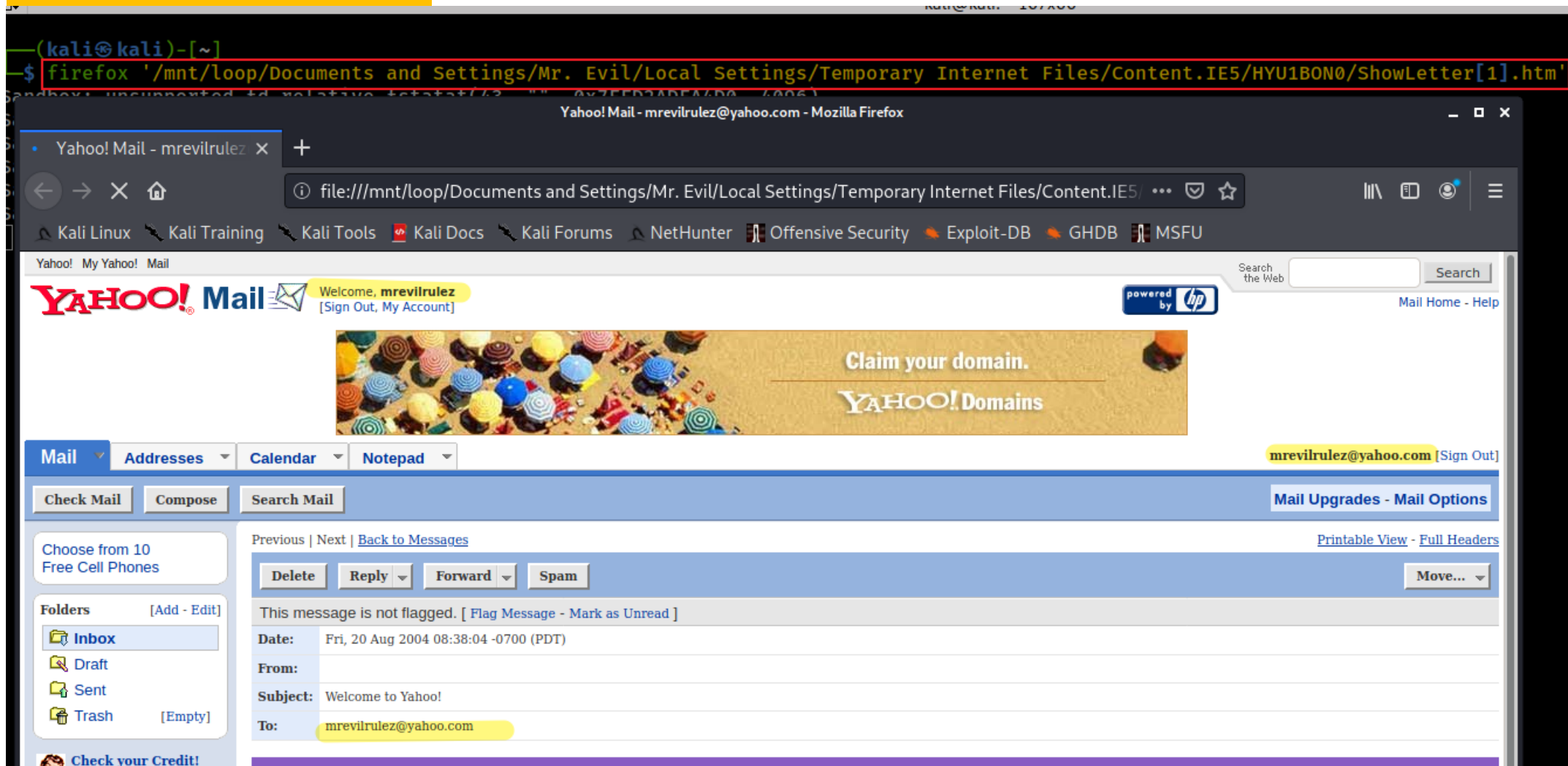
reverse the result of comparisons

27. Yahoo mail saves copies (web cache) of the email under what file name?

Search email under Mr. Evil's account

```
(kali㉿kali)-[~]
$ grep -ir 'mrevilrulez@yahoo.com' '/mnt/loop/Documents and Settings/Mr. Evil/'
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/ShowFolder[1].htm:Yahoo! Mail - mrevilrulez@yahoo.com</title>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/ShowFolder[1].htm: <b>mrevilrulez@yahoo.co
m</b> [a href="/ym/Logout?YY=60138&.first=1&inc=25&order=down&sort=date&pos=0&view=&head=&box=Inbox&YY=60138">Sign Out</a>]
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/ShowLetter[1].htm:Yahoo! Mail - mrevilrulez@yahoo.com</title>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/ShowLetter[1].htm: <b>mrevilrulez@yahoo.co
m</b> [a href="/ym/Logout?YY=90802&.first=1&order=down&sort=date&pos=0&YY=90802">Sign Out</a>]
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/ShowLetter[1].htm:<tr><td class=label nowrap>To:</td><td>mrevilr
ulez@yahoo.com</td></tr>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/ShowLetter[1].htm: <br>Dear mrevilrulez@yahoo.com,<br><br>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/ShowLetter[1].htm: Welcome to Yahoo! Mail, a smarter way o
f keeping in touch. With a whopping <i>100MB of email storage, message size up to 10MB, and great virus and spam protection</i>, it's hard to believe it's <i>free!</i>
Start using your new address right away: <b>mrevilrulez@yahoo.com</b></font>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/last[1].htm:<tr><td colspan=2 align=left><font face="Arial" size
=-1 color="#646464"><b>Your New Yahoo! Mail Address: <font color="#000000">mrevilrulez@yahoo.com</font></b></font></td></tr>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/login[1].htm:Yahoo! Mail - mrevilrulez@yahoo.com</title>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/HYU1BON0/login[1].htm: <b>mrevilrulez@yahoo.com</b> [a
href="/ym/Logout?YY=78169&.first=1&YY=78169">Sign Out</a>]
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/PN0J70QM/ShowLetter[1].htm:Yahoo! Mail - mrevilrulez@yahoo.com</title>
/mnt/loop/Documents and Settings/Mr. Evil/Local Settings/Temporary Internet Files/Content.IE5/PN0J70QM/ShowLetter[1].htm: <b>mrevilrulez@yahoo.co
m</b> [a href="/ym/Logout?YY=27630&.first=1&inc=25&order=down&sort=date&pos=0&view=&head=&box=Inbox&YY=27630">Sign Out</a>]
```

View cached webpages



28. How many executable files are in the recycle bin?

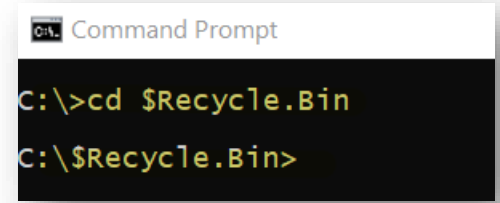
```
kali@kali: ~ 128x44  
  
(kali㉿kali)-[~]  
$ ls -l '/mnt/loop/RECYCLER/S-1-5-21-2000478354-688789844-1708537768-1003'  
total 12103  
-rwxrwxrwx 1 root root 2160043 Aug 25 2004 Dc1.exe  
-rwxrwxrwx 1 root root 1324940 Aug 27 2004 Dc2.exe  
-rwxrwxrwx 1 root root 442417 Aug 27 2004 Dc3.exe  
-rwxrwxrwx 1 root root 8460502 Aug 27 2004 Dc4.exe  
-rwxrwxrwx 1 root root 65 Aug 25 2004 desktop.ini  
-rwxrwxrwx 1 root root 3220 Aug 27 2004 INFO2
```

29. Are these files really deleted?

- No. They can be restored.
- We will try to recover the deleted files' information

Recover deleted files' information

- Extract deleted file from Windows Recycle Bin
 - using Rifiuti2
- Rifiuti2 is a for analyzing Windows Recycle Bin INFO2 file.
 - The complete path and file or folder name is stored in a hidden file called INFO2 which is inside the Recycled or Recycler folder.
- Rifiuti2 can extract
 - file deletion time,
 - original path
 - size of deleted files ad
 - whether the trashed files have been permanently removed.



```
Command Prompt
C:\>cd $Recycle.Bin
C:\$Recycle.Bin>
```

rifiuti2 features

- Handles oldest (Win95) to newest (Win 10 and Server 2019) recycle bin format
- Windows 95 – 2003 uses a single index file named INFO or INFO2
- Vista or above uses one index file for each deleted item
- 64-bit file size support
- Supports all localized versions of Windows — both Unicode-based ones and legacy ones (using ANSI code page)
- Supports output in XML format as well as original tab-delimited text
- Obscure features such as recycle bin on network share (\\server\share)

Install rifiuti2

Install using apt

```
(kali㉿kali)-[~]  
$ sudo apt-get install rifiuti2  
[sudo] password for kali:  
Reading package lists... Done  
Building dependency tree
```

Check the version

```
(kali㉿kali)-[~]  
$ rifiuti2 -v  
rifiuti2 0.7.0  
rifiuti2 is distributed under the BSD 3-Clause License.  
Information about rifiuti2 can be found on  
  
https://abelcheung.github.io/rifiuti2/
```

Show Recycler Bin

```
kali@kali: ~ 128x44  
(kali㉿kali)-[~]  
$ ls -l '/mnt/loop/RECYCLER/'  
total 4  
drwxrwxrwx 1 root root 4096 Aug 27 2004 S-1-5-21-2000478354-688789844-1708537768-1003
```

sid of Mr. Evil

Show INFO2 in Recycler Bin

```
(kali㉿kali)-[~]  
$ ls -l '/mnt/loop/RECYCLER/S-1-5-21-2000478354-688789844-1708537768-1003'  
total 12103  
-rwxrwxrwx 1 root root 2160043 Aug 25 2004 Dc1.exe  
-rwxrwxrwx 1 root root 1324940 Aug 27 2004 Dc2.exe  
-rwxrwxrwx 1 root root 442417 Aug 27 2004 Dc3.exe  
-rwxrwxrwx 1 root root 8460502 Aug 27 2004 Dc4.exe  
-rwxrwxrwx 1 root root 65 Aug 25 2004 desktop.ini  
-rwxrwxrwx 1 root root 3220 Aug 27 2004 INFO2
```

```
(kali@kali)-[~]
```

```
$ rifiuti2 --help
```

Usage:

```
rifiuti2 [OPTION...] INFO2
```

Parse INFO2 file and dump recycle bin data.

Help Options:

-h, --help	Show help options
--help-all	Show all help options
--help-text	Show plain text output options

Application Options:

-o, --output=FILE	Write output to FILE
-x, --xml	Output in XML format instead of tab-delimited values
-z, --localtime	Present deletion time in time zone of local system (default is UTC)
-v, --version	Print version information and exit
-l, --legacy-filename=CODEPAGE	Show legacy (8.3) path if available and specify its CODEPAGE

Show the full path of deleted files

kali@kali: ~ 128x44

(kali@kali)-[~]

\$ rifiuti2 '/mnt/loop/RECYCLER/S-1-5-21-2000478354-688789844-1708537768-1003/INFO2'

Recycle bin path: '/mnt/loop/RECYCLER/S-1-5-21-2000478354-688789844-1708537768-1003/INFO2'

Version: 5

OS Guess: Windows XP or 2003

Time zone: Coordinated Universal Time (UTC) [+0000]

Index	Deleted Time	Gone?	Size	Path
1	2004-08-25 16:18:25	No	2160128	C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe
2	2004-08-27 15:12:30	No	1325056	C:\Documents and Settings\Mr. Evil\Desktop\netstumblerinstaller_0_4_0.exe
3	2004-08-27 15:15:26	No	442880	C:\Documents and Settings\Mr. Evil\Desktop\WinPcap_3_01_a.exe
4	2004-08-27 15:29:58	No	8460800	C:\Documents and Settings\Mr. Evil\Desktop\ethereal-setup-0.10.6.exe

30. How many files are actually reported to be deleted by the file system?

Show deleted files (365)

```
(kali@kali)-[~/hacking_case]
$ fls -rFd -o 63 SCHARDT.dd | wc -l
365
    -d: deleted file
```

31. Perform an Anti-Virus check. Are there any viruses on the computer?

- Suspect may produce virus/be infected.
- Use Clam AntiVirus to scan Windows
- ClamAV is a free software, cross-platform and open-source antivirus software toolkit able to detect many types of malicious software, including viruses.
- One of its main uses is on mail servers as a server-side email virus scanner.

Installation and verification

```
kali@kali: ~/hack
(kaliⓈkali)-[~/hacking_case]
$ sudo apt-get install clamav
[sudo] password for kali:
Reading package lists... Done
Building dependency tree
```

```
(kaliⓈkali)-[~/hacking_case]
$ clamscan --help

Clam AntiVirus: Scanner 0.103.0
By The ClamAV Team: https://www.clamav.net/about.html#credits
(C) 2020 Cisco Systems, Inc.
```

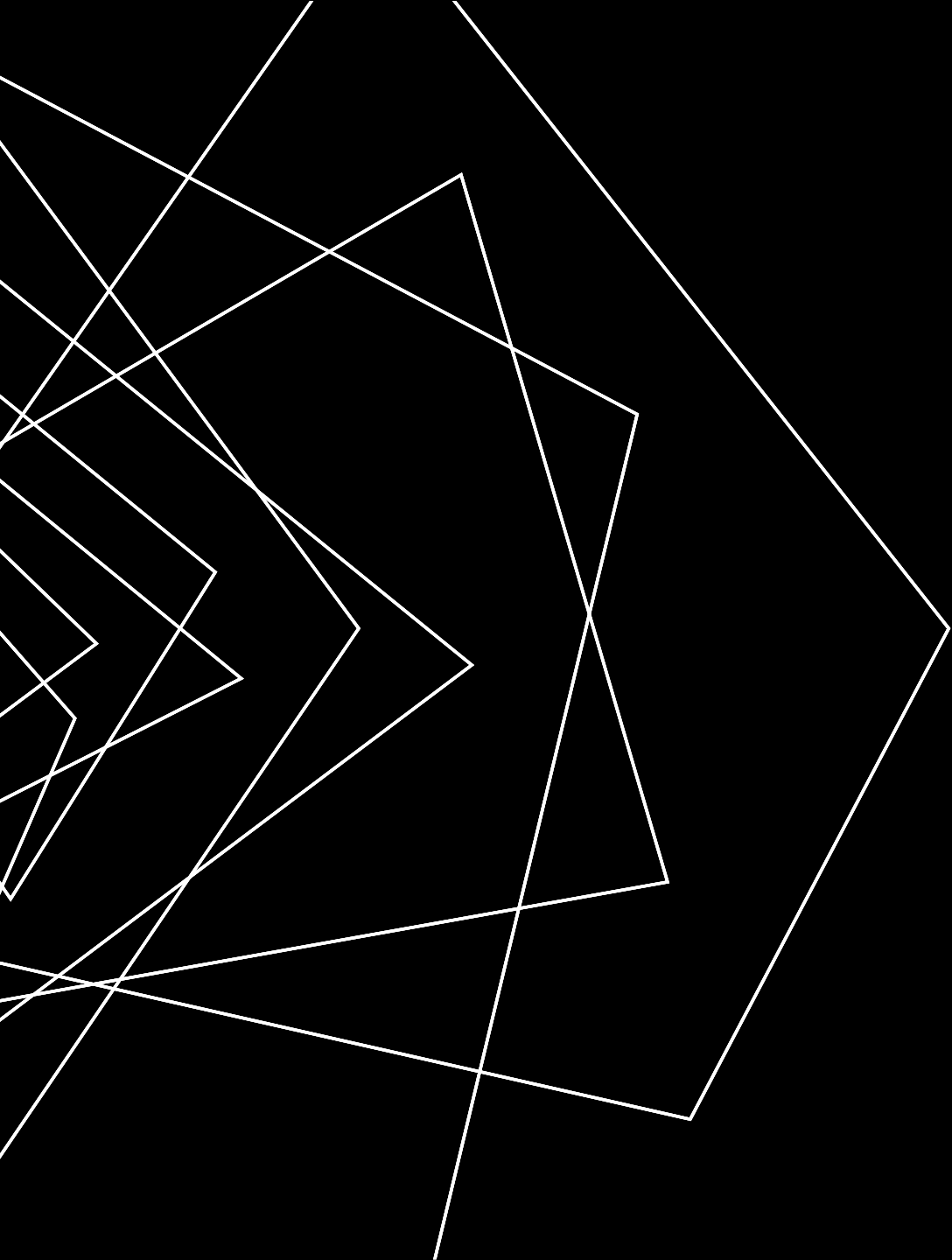
Scan results

```
kali@kali: ~/hacking_case 128x39
(kali@kali)-[~/hacking_case]
$ clamscan -r -i '/mnt/loop'
/mnt/loop/My Documents/COMMANDS/enum.exe: Win.Tool.EnumPlus-1 FOUND
/mnt/loop/My Documents/COMMANDS/SAMDUMP.EXE: Win.Trojan.Pwdump-2 FOUND
/mnt/loop/My Documents/COMMANDS/snitch.exe: Win.Trojan.Snitch-1 FOUND
/mnt/loop/My Documents/ENUMERATION/NT/enum/enum.tar.gz: Win.Tool.EnumPlus-1 FOUND
/mnt/loop/My Documents/ENUMERATION/NT/enum/files/enum.exe: Win.Tool.EnumPlus-1 FOUND
/mnt/loop/My Documents/ENUMERATION/NT/Legion/Chrono.dl_: Win.Trojan.Bruteforce-3 FOUND
/mnt/loop/My Documents/ENUMERATION/NT/Legion/NetTools.ex_: Win.Trojan.Spion-4 FOUND
/mnt/loop/My Documents/ENUMERATION/NT/ntreskit.zip: Win.Trojan.Nemo-1 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/Brutus/BrutusA2.exe: Win.Tool.Brutus-3 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/brutus.zip: Win.Tool.Brutus-3 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/Get Admin/GetAdmin.exe: Win.Exploit.WinNT-3 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/lsadump2/lsadump2.exe: Win.Trojan.Lsadump-1 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/lsadump2/lsadump2.zip: Win.Trojan.Lsadump-1 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/netbus/NetBus170.zip: Win.Trojan.Netbus-2 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/sechole/SECHOLE.EXE: Win.Trojan.Sehole-1 FOUND
/mnt/loop/My Documents/EXPLOITATION/NT/sechole/sechole3.zip: Win.Trojan.Sehole-1 FOUND
/mnt/loop/My Documents/FOOTPRINTING/NT/superscan/superscan.exe: Win.Trojan.Agent-6240252-0 FOUND
/mnt/loop/Program Files/Cain/Abel.dll: Win.Trojan.Cain-9 FOUND
/mnt/loop/Program Files/Online Services/MSN50/MSN50.CAB: Txt.Malware.CMSTPEvasion-6664831-0 FOUND
/mnt/loop/WIN98/WIN98_OL.CAB: Txt.Malware.CMSTPEvasion-6664831-0 FOUND
/mnt/loop/WINDOWS/system32/ahui.exe: Win.Virus.Virut-6804272-0 FOUND
/mnt/loop/WINDOWS/system32/dllcache/ahui.exe: Win.Virus.Virut-6804272-0 FOUND

----- SCAN SUMMARY -----
Known viruses: 8559038
Engine version: 0.103.0
Scanned directories: 766
Scanned files: 11305
Infected files: 22
Data scanned: 1983.00 MB
Data read: 1768.03 MB (ratio 1.12:1)
Time: 1512.949 sec (25 m 12 s)
Start Date: 2021:02:22 19:27:58
End Date: 2021:02:22 19:53:11
```

What is your conclusion?

- Is Greg Schardt a hacker?



Thank you

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